

$$\begin{cases} \text{div } \vec{H} = 0 \\ \text{rot } \vec{E} = -\frac{1}{c} \frac{\partial \vec{H}}{\partial t} \end{cases} \quad \frac{\partial \tilde{F}^{\mu\nu}}{\partial x^\mu} = 0$$

1 Действие для ИМ поля

$$S_f = \int c dt \int d^3 \vec{r} (a F^{\mu\nu} F_{\mu\nu} + b \tilde{F}^{\mu\nu} F_{\mu\nu})$$

$\int d^3 \vec{r}$ - 4-объем, рел. инвар.

$$\tilde{F}^{\mu\nu} F_{\mu\nu} = -4(\vec{H}, \vec{E}) - \text{псевдоскаляр}$$

$\sim 4 \epsilon^{\mu\nu\alpha\beta} \partial_\mu (A_\nu \partial_\alpha A_\beta)$ $\Rightarrow$ положим $b=0$

$$F^{\mu\nu} F_{\mu\nu} = 2(\vec{H}^2 - \vec{E}^2) \sim -\frac{1}{c^2} \left(\frac{\partial \vec{A}}{\partial t} \right)^2 \quad a < 0$$

$$a = -\frac{1}{16\pi c} \quad (\text{в СГС})$$

$$S_f = -\frac{1}{16\pi c} \int d^4 x F^{\mu\nu} F_{\mu\nu} = \int dt \int d^3 \vec{r} \frac{\vec{E}^2 - \vec{H}^2}{8\pi} \equiv \int dt \int d^3 \vec{r} L$$

$$L = \frac{\vec{E}^2 - \vec{H}^2}{8\pi} \quad \text{пл-ть лагранжиана ИМ поля}$$

$$\int d^3 \vec{r} \frac{\vec{E}^2 - \vec{H}^2}{8\pi} \quad \text{лагранжиан ИМ поля}$$

полное действие частиц и поля

$$S = S_p + S_{p-1} + S_f = -\sum_i m_i c \int dS_i - \underbrace{\sum_i \frac{e_i}{c} \int d^4 x_i^\mu A_\mu(x_i^\mu)}_{(cdt, d\vec{r}_i) \quad (\varphi, A)} - \frac{1}{16\pi c} \int d^4 x F^{\mu\nu} F_{\mu\nu}$$

$$= -\sum_i e_i \int dt \varphi(\vec{r}_i) + \sum_i \frac{e_i}{c} \int dt \vec{A}(\vec{r}_i) \vec{v}_i$$

2 Плотности заряда, пл-ть тока, пл-ть 4-тока

$$\rho(t, \vec{r}) = \sum_i e_i \delta(\vec{r} - \vec{r}_i(t)) \quad \text{пл-ть заряда}$$

$$\int d^3 \vec{r} \delta(\vec{r} - \vec{r}_i) = 1$$

$$\vec{j}(t, \vec{r}) = \sum_i e_i \vec{v}_i(t) \delta(\vec{r} - \vec{r}_i(t)) \quad \text{пл-ть тока}$$

$$\underbrace{d^4 x^\mu}_{4\text{-б-р}} = g dV \underbrace{\frac{dx^\mu}{dt}}_{4\text{-б-р}} dt = g \frac{dx^\mu}{dt} \underbrace{\frac{1}{c} d^3 \vec{r}}_{\text{скаляр}}$$

$$j^\mu = g \frac{dx^\mu}{dt} = (cg, g\vec{v}) = (cg, \vec{j}) \quad \text{4-б-р плотности тока (4-ток)}$$

$$S_{p-1} = -\frac{1}{c^2} \int c dt \int d^3 \vec{r} \varphi(\vec{r}) \sum_i e_i \delta(\vec{r} - \vec{r}_i(t)) c + \frac{1}{c^2} \int c dt \int d^3 \vec{r} \vec{A}(\vec{r}) \sum_i e_i \vec{v}_i(t) \delta(\vec{r} - \vec{r}_i(t)) c =$$

$$= -\frac{1}{c^2} \int d^4 x g \varphi + \frac{1}{c^2} \int d^4 x \vec{A} \vec{j} = -\frac{1}{c^2} \int d^4 x A^\mu j_\mu$$

$$S_{p-1} + S_f = -\frac{1}{c^2} \int d^4 x A^\mu j_\mu - \frac{1}{16\pi c} \int d^4 x F^{\mu\nu} F_{\mu\nu}$$

3 2 пара ур-н Максвелла

(движение частиц задано, варьирруем перемен. поля)

$$\delta S = \delta S_{p-1} + \delta S_f = -\frac{1}{c^2} \int dV j^\mu \delta A_\mu - \frac{1}{8\pi c} \int dV F^{\mu\nu} \delta F_{\mu\nu}$$

$$F^{\mu\nu} \delta F_{\mu\nu} = F^{\mu\nu} \delta \left(\frac{\partial A_\nu}{\partial x^\mu} - \frac{\partial A_\mu}{\partial x^\nu} \right) = \underbrace{F^{\mu\nu} \frac{\partial}{\partial x^\mu} \delta A_\nu}_{F^{\mu\nu} \frac{\partial}{\partial x^\mu} \delta A_\mu} - F^{\mu\nu} \frac{\partial}{\partial x^\nu} \delta A_\mu = -2 F^{\mu\nu} \frac{\partial}{\partial x^\nu} \delta A_\mu$$

$$\delta S = -\frac{1}{c^2} \int dV j^\mu \delta A_\mu + \frac{1}{4\pi c} \int dV F^{\mu\nu} \frac{\partial}{\partial x^\nu} \delta A_\mu = -\frac{1}{c^2} \int dV j^\mu \delta A_\mu - \frac{1}{4\pi c} \int dV \left(\frac{\partial}{\partial x^\nu} F^{\mu\nu} \right) \delta A_\mu + \frac{1}{4\pi c} \oint dS_\nu F^{\mu\nu} \delta A_\mu = -\frac{1}{4\pi c} \int dV \left(\frac{\partial}{\partial x^\nu} F^{\mu\nu} + \frac{4\pi}{c} j^\mu \right) \delta A_\mu + \frac{1}{4\pi c} \oint dS_\nu F^{\mu\nu} \delta A_\mu$$

↑
нов-ые в пр-ве Макс.

↑
поля у. на беск.

$$\delta S = 0 \quad \forall \delta A_\mu = 0$$

$$\frac{\partial F^{\mu\nu}}{\partial x^\nu} = -\frac{4\pi}{c} j^\mu$$

2-я пара ур-й Максвелла

$$\mu=0$$

$$F^{0j} = -(\bar{E})_j$$

$$\text{div } \bar{E} = 4\pi g$$

$$\mu=x \quad (\bar{E})_x$$

$$\frac{\partial F^{x,ct}}{\partial (ct)} + \frac{\partial F^{xy}}{\partial y} + \frac{\partial F^{xz}}{\partial z} = -\frac{4\pi}{c} (j)_x$$

$$\frac{1}{c} \frac{\partial E_x}{\partial t} - \frac{\partial H_z}{\partial y} + \frac{\partial H_y}{\partial z} = -\frac{4\pi}{c} j_x$$

- (rot H)_x

$$\text{rot } \bar{H} = \frac{1}{c} \frac{\partial \bar{E}}{\partial t} + \frac{4\pi}{c} \bar{j}$$

$$\oint d\bar{S} \bar{E} = 4\pi Q$$

$$\oint d\bar{L} \bar{H} = \frac{1}{c} \frac{\partial}{\partial t} \oint d\bar{S} \bar{E} + \frac{4\pi}{c} \int d\bar{S} \bar{j}$$

3-я пара Максвелла

$$4\text{D:} \quad \frac{\partial \hat{F}^{\mu\nu}}{\partial x^\nu} = 0 \quad \text{1 пара}$$

$$\frac{\partial F^{\mu\nu}}{\partial x^\nu} = -\frac{4\pi}{c} j^\mu \quad \text{2 пара}$$

$$3\text{D:} \quad \begin{cases} \text{div } \bar{H} = 0 \\ \text{rot } \bar{E} = -\frac{1}{c} \frac{\partial \bar{H}}{\partial t} \end{cases}$$

$$\begin{cases} \text{div } \bar{E} = 4\pi g \\ \text{rot } \bar{H} = \frac{1}{c} \frac{\partial \bar{E}}{\partial t} + \frac{4\pi}{c} \bar{j} \end{cases}$$

$\frac{4\pi}{c} \left(\bar{j} + \frac{1}{4\pi} \frac{\partial \bar{E}}{\partial t} \right)$

инт. форма:

$$\begin{cases} \oint \bar{H} d\bar{S} = 0 \\ \oint d\bar{L} \bar{E} = -\frac{1}{c} \frac{\partial}{\partial t} \oint d\bar{S} \bar{H} \\ \oint \bar{E} d\bar{S} = 4\pi Q \\ \oint d\bar{L} \bar{H} = \frac{1}{c} \frac{\partial}{\partial t} \oint d\bar{S} \bar{E} + \frac{4\pi}{c} \int d\bar{S} \bar{j} \end{cases}$$

4 Закон сохр. эл. заряда

$$\frac{\partial F^{\mu\nu}}{\partial x^\nu} = -\frac{4\pi}{c} j^\mu$$

$$\frac{\partial^2 F^{\mu\nu}}{\partial x^\mu \partial x^\nu} = -\frac{4\pi}{c} \frac{\partial j^\mu}{\partial x^\mu} \Rightarrow \frac{\partial j^\mu}{\partial x^\mu} = 0$$

0 (вертикаль симм. и антисимм.)

$$\frac{\partial g}{\partial t} + \text{div } j = 0$$

$$\oint j d\bar{S} = -\frac{\partial Q}{\partial t}$$

5 Плотность энергии частиц и поля

$$(\bar{H}, \cdot) \quad \text{rot} \bar{E} = -\frac{1}{c} \frac{\partial \bar{H}}{\partial t}$$

$$(\bar{E}, \cdot) \quad \text{rot} \bar{H} = \frac{1}{c} \frac{\partial \bar{E}}{\partial t} + \frac{4\pi}{c} \bar{j}$$

$$\Rightarrow (\bar{H}, \text{rot} \bar{E}) - (\bar{E}, \text{rot} \bar{H}) = -\frac{1}{c} (\bar{H}, \frac{\partial \bar{H}}{\partial t}) - \frac{1}{c} (\bar{E}, \frac{\partial \bar{E}}{\partial t}) - \frac{4\pi}{c} (\bar{j}, \bar{E})$$

$$\text{div} [\bar{E}, \bar{H}] + \frac{1}{c} \frac{\partial}{\partial t} \left(\frac{\bar{E}^2 + \bar{H}^2}{2} \right) + \frac{4\pi}{c} (\bar{j}, \bar{E}) = 0$$

$$\frac{\partial}{\partial t} \left(\frac{\bar{E}^2 + \bar{H}^2}{8\pi} \right) + \text{div} \left(\frac{c}{4\pi} [\bar{E}, \bar{H}] \right) + (\bar{j}, \bar{E}) = 0 \quad (*)$$

$$w = \frac{\bar{E}^2 + \bar{H}^2}{8\pi}, \quad \bar{S} = \frac{c}{4\pi} [\bar{E}, \bar{H}]$$

умм. (*) по всему пр-ву $\sum_i e_i \bar{V}_i(t) \delta(\bar{r} - \bar{r}_i(t))$

$$\frac{\partial}{\partial t} \int d^3 \bar{r} w + \int d^3 \bar{r} \text{div} \bar{S} + \int d^3 \bar{r} (\bar{j}, \bar{E}) = 0$$

$$\oint d\bar{S} \bar{S} = 0$$

эл-т пов-тия

$$\sum_i e_i (\bar{V}_i(t), \bar{E}(t, \bar{r}_i)) = \frac{d}{dt} \xi_{\text{кин}}$$

$$\frac{d}{dt} \left(\int d^3 \bar{r} w + \xi_{\text{кин}} \right) = 0$$

$$\Rightarrow \int d^3 \bar{r} w \quad \text{полная энергия поля}$$

w м-ть энергии Эл поля

Умм (*) по нулевой объёму (j=0)

$$\frac{\partial}{\partial t} \left(\int_V d^3 \bar{r} w \right) = - \oint d\bar{S} \bar{S}$$

$$\Rightarrow \bar{S} \quad \text{м-ть потока энергии (в-р Пойнтинга)}$$

$$\frac{\partial w}{\partial t} + \text{div} \bar{S} + (\bar{j}, \bar{E}) = 0 \quad \text{баланс энергии частиц и поля}$$

6 Баланс импульса частиц и поля

$$[\bar{E}, \cdot] \quad \text{rot} \bar{E} = -\frac{1}{c} \frac{\partial \bar{H}}{\partial t}$$

$$[\bar{H}, \cdot] \quad \text{rot} \bar{H} = \frac{1}{c} \frac{\partial \bar{E}}{\partial t} + \frac{4\pi}{c} \bar{j} - \frac{\partial}{\partial t} \left(\frac{1}{c} [\bar{E}, \bar{H}] \right)$$

$$[\bar{E}, \text{rot} \bar{E}] + [\bar{H}, \text{rot} \bar{H}] = -\frac{1}{c} [\bar{E}, \frac{\partial \bar{H}}{\partial t}] + \frac{1}{c} [\bar{H}, \frac{\partial \bar{E}}{\partial t}] + \frac{4\pi}{c} [\bar{H}, \bar{j}]$$

$$[\bar{E}, \text{rot} \bar{E}]_i = \epsilon_{ijk} E_j \epsilon_{k\ell m} \partial_\ell E_m = (\delta_{i\ell} \delta_{jm} - \delta_{im} \delta_{j\ell}) E_j \partial_\ell E_m = E_m \partial_i E_m -$$

$$- E_\ell \partial_\ell E_i = \frac{1}{2} \partial_i (\bar{E}^2) - \partial_\ell (E_i E_\ell) + E_i \partial_\ell E_\ell = \partial_\ell (\delta_{i\ell} \frac{\bar{E}^2}{2} - E_i E_\ell) + E_i \text{div} \bar{E} = \partial_\ell (\delta_{i\ell} \frac{\bar{E}^2}{2} - E_i E_\ell) + E_i 4\pi g$$

$$[\bar{H}, \text{rot} \bar{H}]_i = \partial_\ell (\delta_{i\ell} \frac{\bar{H}^2}{2} - H_i H_\ell) + H_i \text{div} \bar{H} = \partial_\ell (\delta_{i\ell} \frac{\bar{H}^2}{2} - H_i H_\ell)$$

$$\partial_\ell \left(\delta_{i\ell} \frac{\bar{E}^2 + \bar{H}^2}{8\pi} - \frac{E_i E_\ell + H_i H_\ell}{4\pi} \right) + \frac{\partial}{\partial t} \left(\frac{S_i}{c^2} \right) + E_i g + \frac{1}{c} [\bar{j}, \bar{H}]_i = 0$$

$$\sigma_{i\ell} = \delta_{i\ell} w - \frac{E_i E_\ell + H_i H_\ell}{4\pi}$$

тензор напряжений Максвелла

$$\frac{\partial}{\partial t} \left(\frac{S_i}{c^2} \right) + \frac{\partial}{\partial x_\ell} \sigma_{i\ell} + E_i g + \frac{1}{c} [\bar{j}, \bar{H}]_i = 0$$